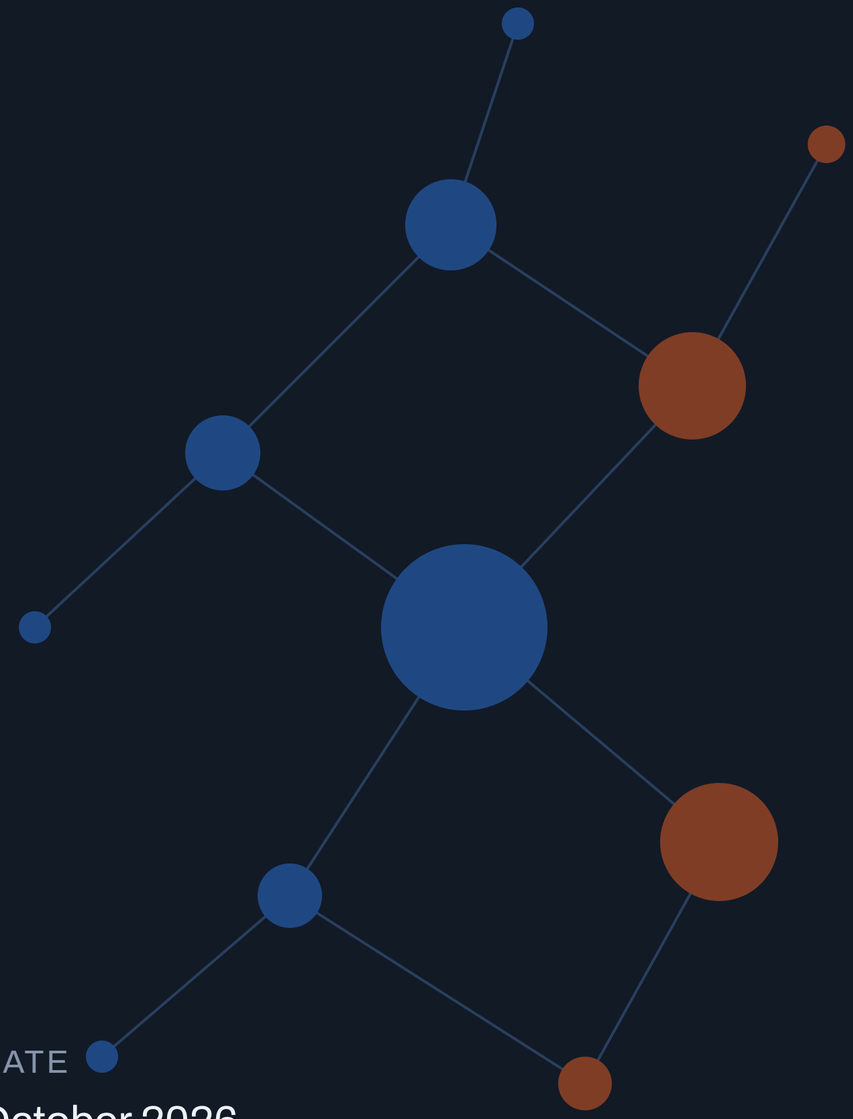
CSE4039 · ADVANCED TEXT & SOCIAL MEDIA ANALYTICS

PageRank on three networks

A from-scratch power-iteration implementation, verified against NetworkX and applied to a social network and a 100K-page Wikipedia link graph



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INSTITUTION

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DATE

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OBJECTIVE

Which pages matter most?

Implement

PageRank by power iteration, from scratch: a readable dict version and a SciPy sparse version.

Verify

Compare every result with NetworkX's nx.pagerank: value gaps, rank correlation, top-10 overlap.

Apply

Rank a 34-member social network and a 100K-page, 8.57M-link Wikipedia graph.

Analyse

Test the damping factor, compare with other centralities, explain what top pages share.

THE ALGORITHM

A random surfer decides what matters

$$PR(i) = (1 - d) / N + d \cdot \Sigma j \to i \; PR(j) / L(j)$$

d = 0.85

The surfer follows one of the page's L(j) out-links.

1 – d = 0.15

The surfer jumps to any of the N pages at random.

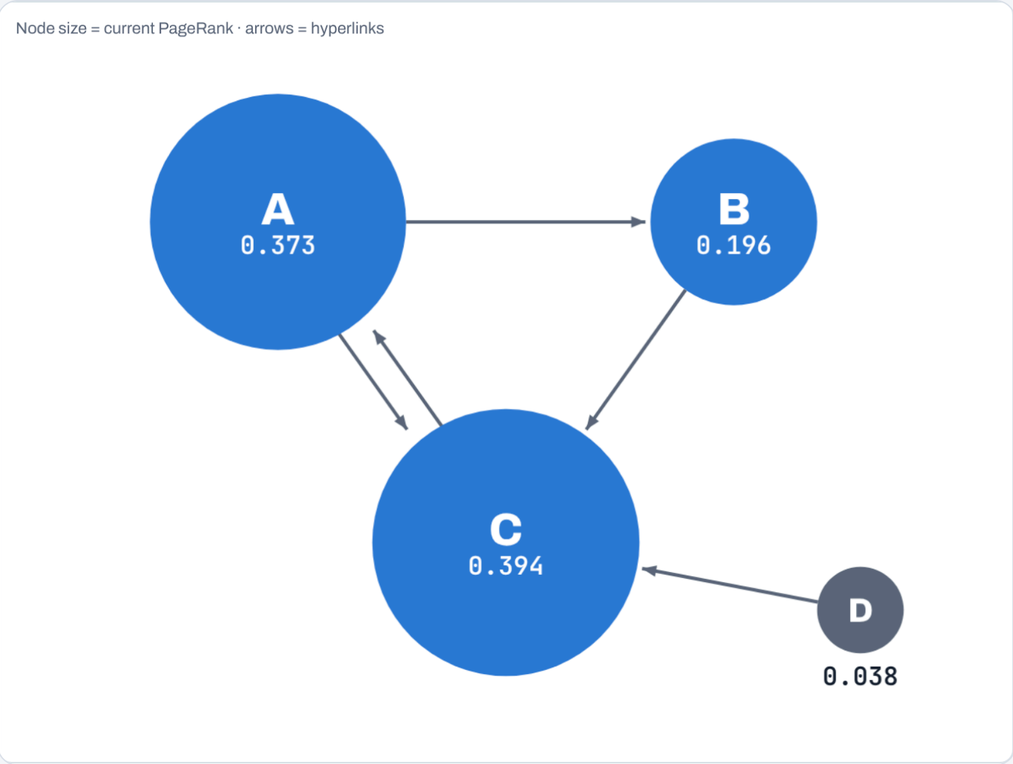
Dangling pages

Pages with no out-links spread their rank evenly.

Power iteration: $r \leftarrow G \cdot r$ until the L1 change between steps is below 10^{-8}.

WORKED EXAMPLE

Four pages, 47 iterations



Page	Links to	PageRank
A	B, C	0.3725
B	C	0.1958
C	A	**0.3941**
D	C	0.0375

C wins: three pages link to it. **A** is second because C gives all its rank to A. **D** has no in-links and keeps only the teleport share (1 – d)/N = 0.0375.

DESIGN / ARCHITECTURE

A five-stage pipeline

1 Data sources	Karate Club, 34 nodes (NetworkX)	Wikipedia link graph 100K (Kaggle)	VIT-AP crawl (robots.txt-aware BFS)
2 Load & clean	Auto-detect edge-list file and columns	Chunked read, page name → integer id	Drop self and duplicate links, close graph
3 PageRank engine	pagerank_simple(): dicts, readable	pagerank_sparse(): SciPy CSR, fast	Dangling mass spread · L1 tol 10^{-8}
4 Verify & analyse	Compare with nx.pagerank	Centrality correlation, HITS	Damping sweep d = 0.5 … 0.95
5 Outputs	CSV tables, PNG charts, results.json	PageRank Explorer web app (JS port)	Slides, demo video, report

IMPLEMENTATION

One algorithm, two engines

	pagerank_simple()	pagerank_sparse()
Graph stored as	dicts of in-link lists	SciPy CSR matrix
One iteration	loop over every in-link	one sparse matrix–vector product
Used on	toy web, Karate Club	Wikipedia, 8.57M links
Memory at 100K pages	pure Python, not used	≈ 100 MB (dense: ≈ 80 GB)
Max gap vs NetworkX	**1.5×10^{-9}**	**1.0×10^{-7}** (20K-page sample)

Both stop when the L1 change between iterations is below 10^{-8} and spread dangling-page rank evenly.

INPUT DATA

Networks of very different size

Karate Club

UNDIRECTED SOCIAL NETWORK

34 members
78 ties

Zachary (1977). NetworkX built-in copy, same as soc-karate on networkrepository.com.

Analysed

Wikipedia 100K

DIRECTED ARTICLE LINKS

99,998 pages
8,572,670 links

Kaggle kutayahin/wikipedia-link-graph-100k, ~28.9M raw links, cleaned and closed.

Analysed

VIT-AP website

DIRECTED HYPERLINKS

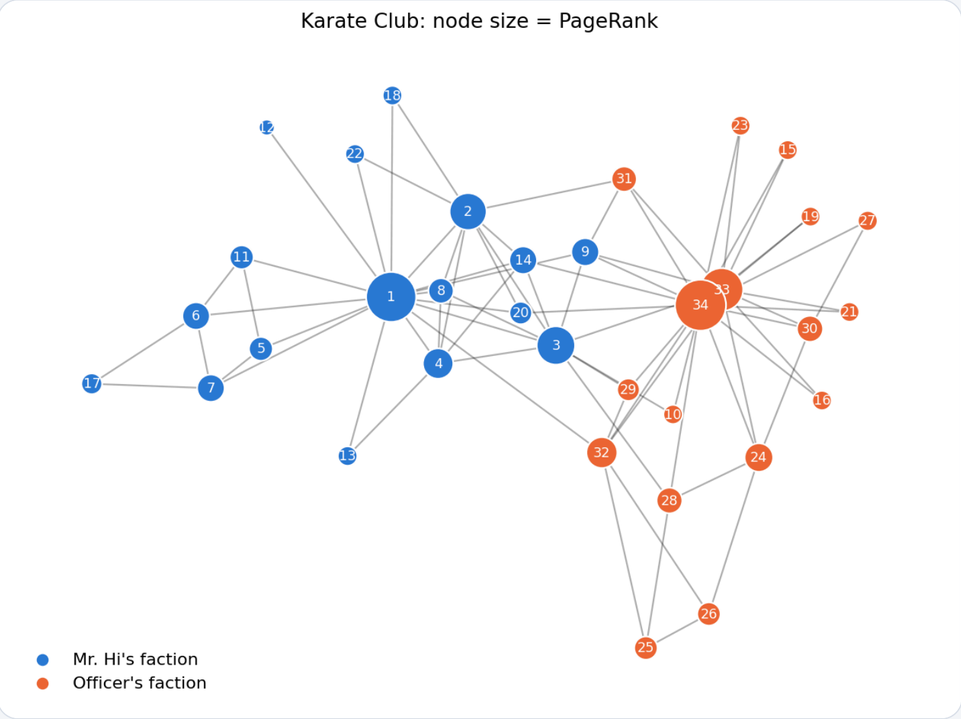
up to 300 pages

Our own polite crawler of vitap.ac.in: obeys robots.txt, waits 0.5 s per request.

Crawler built, run pending

RESULT · KARATE CLUB

The two leaders who split the club rank first



#	Member	Faction	Deg	PR
1	**34** admin	Officer	17	0.1009
2	**1** instructor	Mr. Hi	16	0.0970
3	33	Officer	12	0.0717
4	3	Mr. Hi	10	0.0571
5	2	Mr. Hi	9	0.0529

Node 34 and node 1 together hold **19.8%** of all rank among 34 members.

RESULT · KARATE CLUB

PageRank tracks degree, not exactly

Spearman rank correlation with PageRank (34 members)



Bar length = correlation on a 0 to 1 scale.

Same degree, different rank

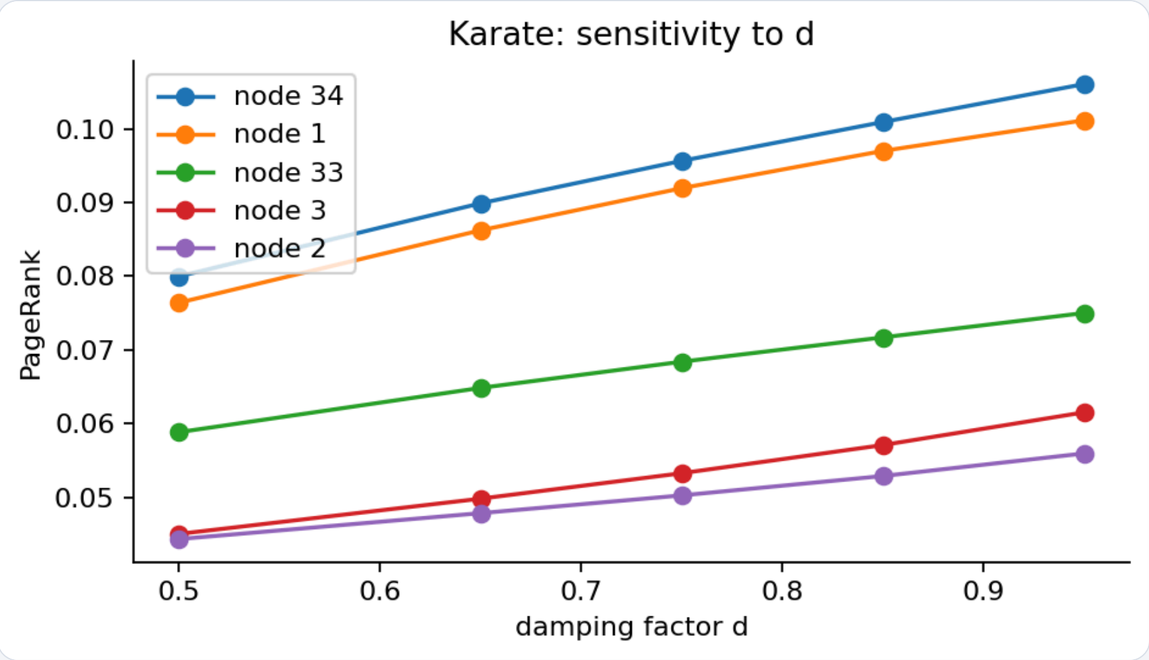
Members 24, 9 and 14 each have 5 friends.

24 → 0.0315
9 → 0.0298
14 → 0.0295

PageRank also counts how important those friends are.

RESULT · DAMPING FACTOR

Higher d: sharper ranks, slower convergence



d	Iter.	PR(34)	Max÷min
0.50	19	0.0800	4.7×
0.65	25	0.0899	6.5×
0.75	32	0.0957	8.2×
0.85	**45**	**0.1009**	**10.6×**
0.95	70	0.1061	14.2×

The top five stay 34, 1, 33, 3, 2 at every d.

VERIFICATION

Our PageRank reproduces NetworkX

1.5×10^{-9}

largest value gap, Karate Club

1.0×10^{-7}

largest value gap, 20K-page Wikipedia sample

1.000

Spearman rank correlation, both graphs

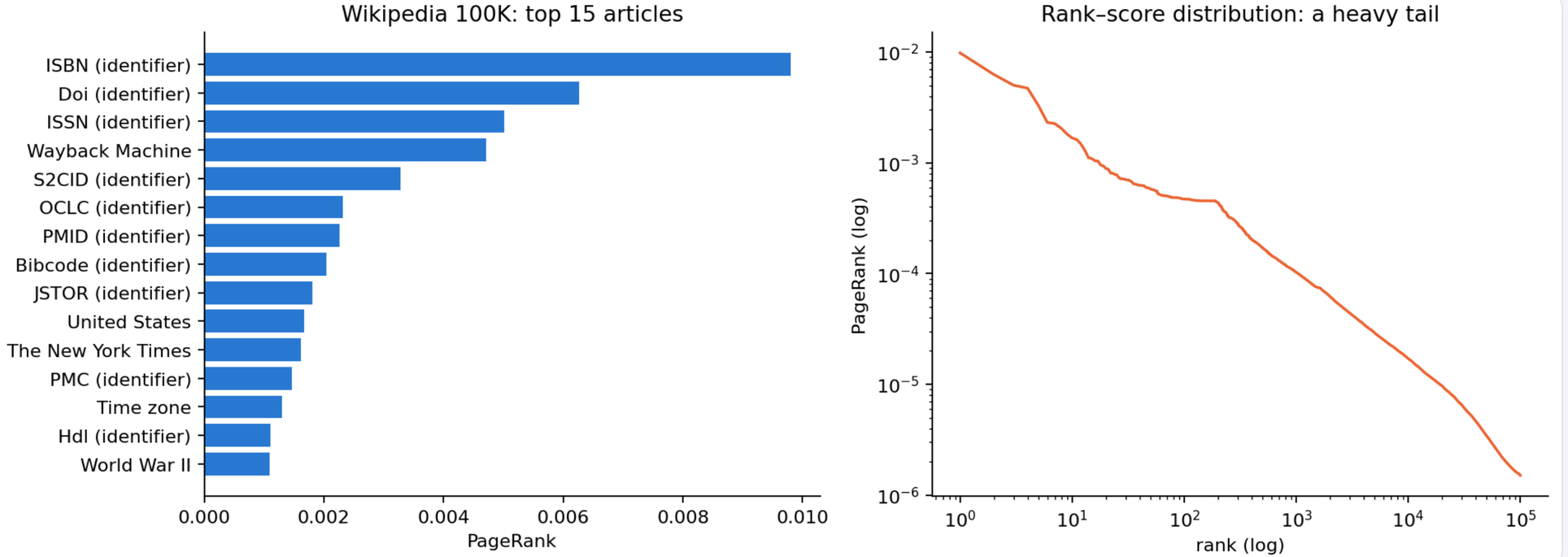
10 / 10

same top-10 pages, both graphs

NetworkX cannot hold the full 8.57M-link graph in memory, so the Wikipedia check uses a random induced subgraph of 20,000 pages.

RESULT · WIKIPEDIA 100K

Rank concentrates in a few hub pages



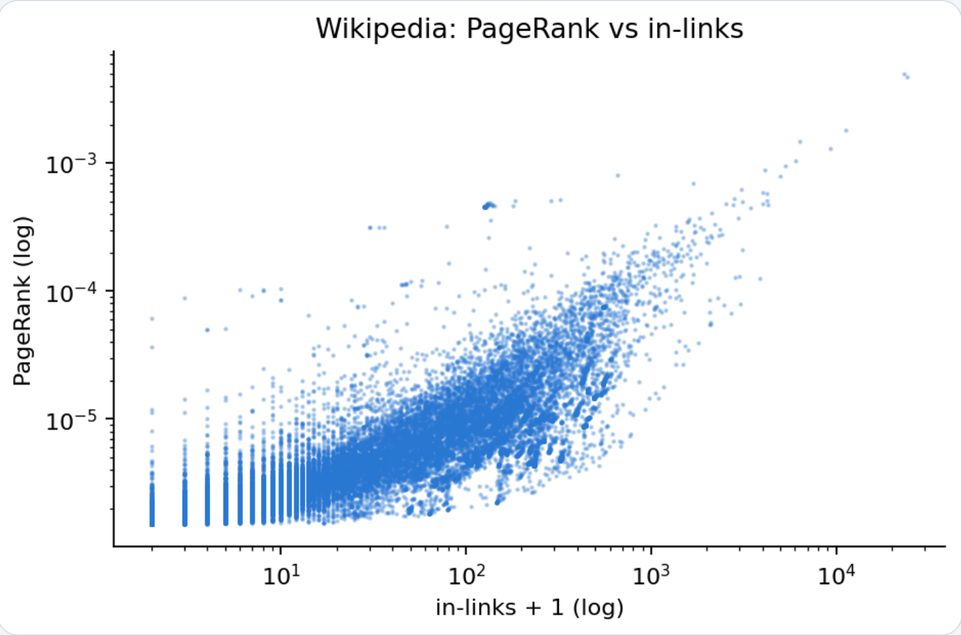
28.2% of all rank sits in the top 1% of pages

61 iterations

203 dangling pages

RESULT · WIKIPEDIA 100K

PageRank weighs who links to you

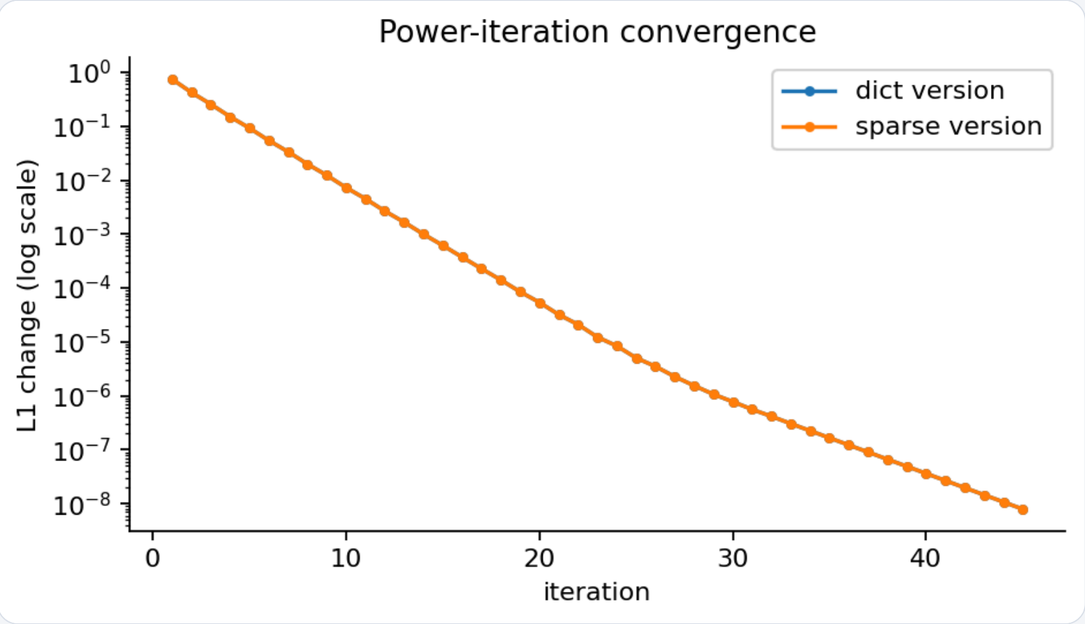


Page	Rank	In-links
ISBN (identifier)	1	51,534
United States	10	9,016
ISO	**16**	**428**
France	20	6,089

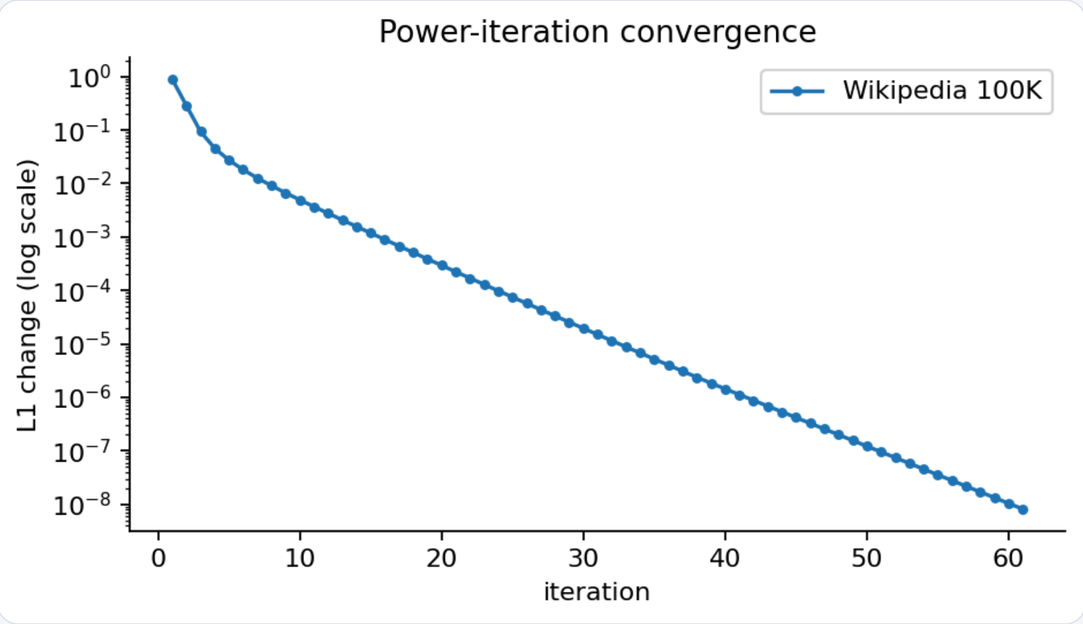
ISO (International Organization for Standardization) has 14× fewer in-links than France yet ranks higher: its links come from ISBN and ISSN.

RESULT · CONVERGENCE

Straight lines on a log scale



Karate Club: 45 iterations

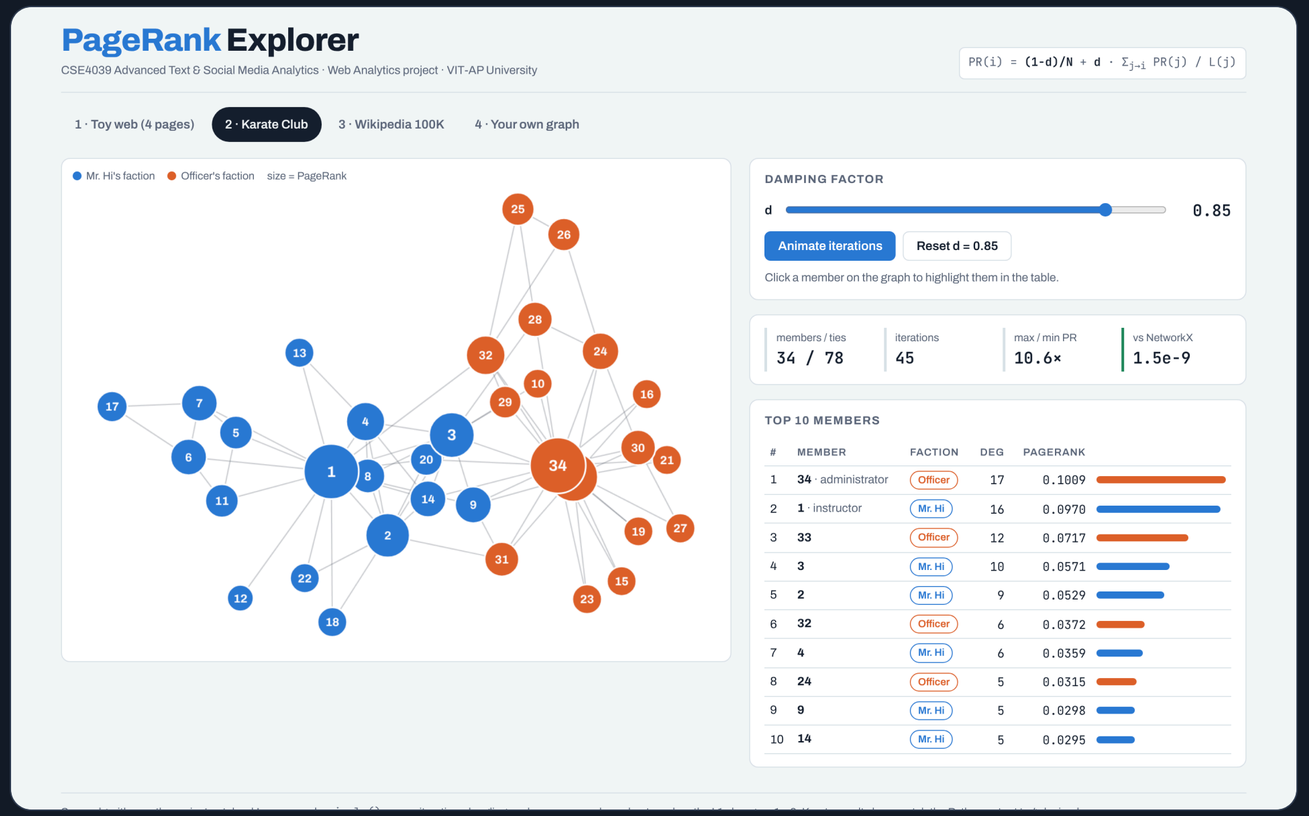


Wikipedia 100K: 61 iterations

The error shrinks by a near-constant factor each step (about 0.67 on Karate Club), never slower than d = 0.85. Stop rule: L1 change below 10^{-8}.

DEMO APPLICATION

PageRank Explorer



The notebook's algorithm ported to JavaScript, with four views:

1 Toy web, one step at a time

2 Karate Club with a d slider

3 Wikipedia 100K results

4 Rank your own edge list

Demo app: GitHub demo-app/ · Demo video: GitHub video/

CONCLUSIONS

What we learned

1. **Correct and fast.** Matches NetworkX to 10^{-9}–10^{-7} and converges in 45–61 iterations.
2. **Finds leaders.** On Karate Club it ranks the two faction leaders first, with no knowledge of the conflict.
3. **Link quality counts.** On Wikipedia, hub identifiers dominate and a few strong in-links beat many weak ones.
4. **d is a trade-off.** Higher damping sharpens ranks but slows convergence; the top five never change.

Limitations

- VIT-AP crawl not run for this submission
- Wikipedia is a closed 100K-page sample
- Karate Club is small and undirected
- The crawler ignores JavaScript-rendered links

THANK YOU · QUESTIONS?

Links and references

SOURCE CODE

github.com/swappy1409/atsma-project

DEMO APPLICATION

…/atsma-project/demo-app

DEMO VIDEO

…/atsma-project/video

FULL REPORT

…/atsma-project/docs

PRESENTATION

…/atsma-project/presentation

REFERENCES

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Zachary, W. W. (1977). An information flow model for conflict and fission in small groups.

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Hagberg, Schult & Swart (2008). NetworkX.